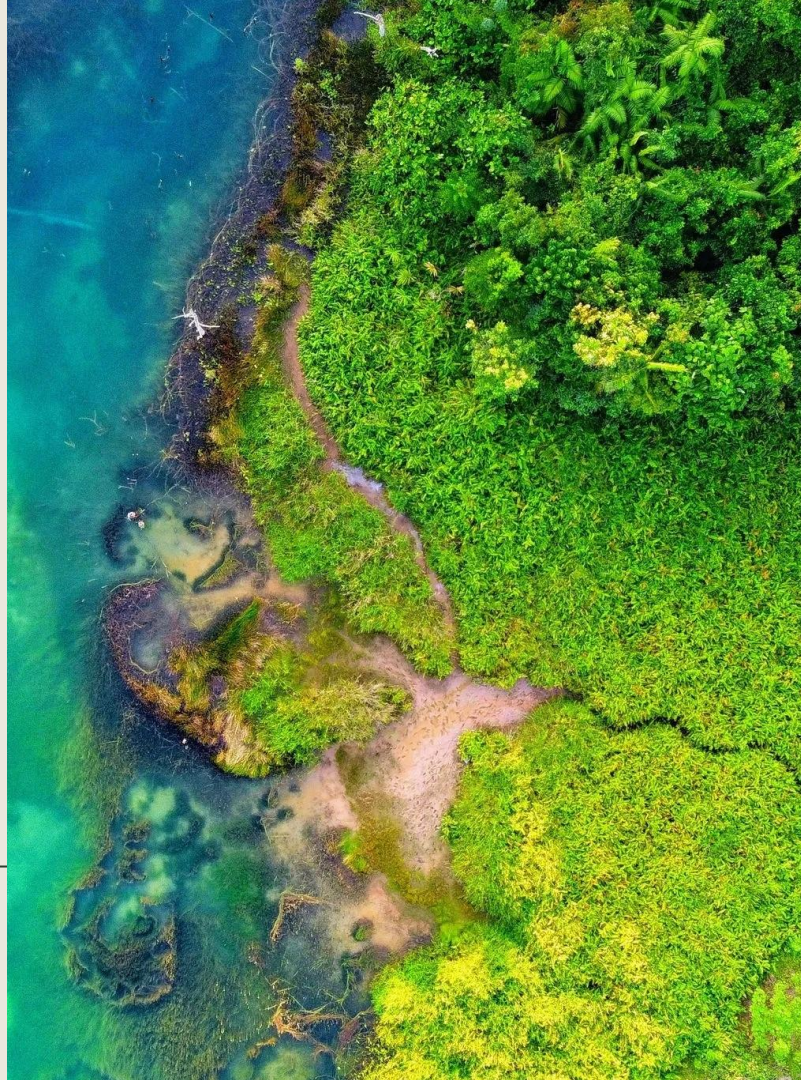
The Main Drivers of Biodiversity Crisis

And How Has Human Activity Restructured Life on Earth?

Zofia Muchowska, Katelyn Keenen, Diana Alvarez Peralta, Matej Dimitrievski, Hesam Owlanj



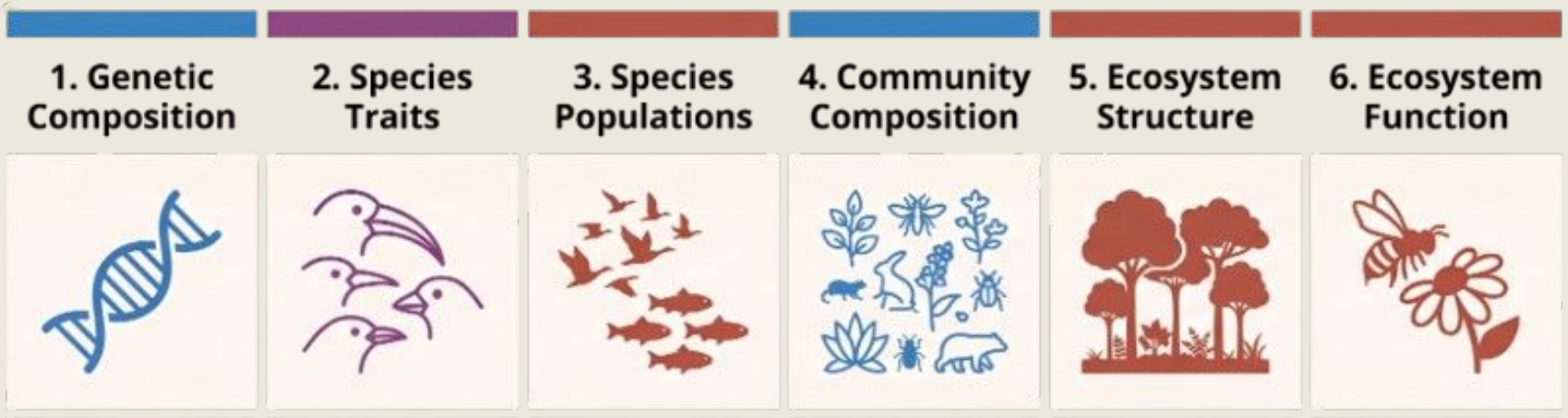
BIODIVERSITY

1. What is it?
2. What is its role?
3. What is the biodiversity crisis?

What are the main drivers of the biodiversity crisis?

1 DIMENSIONS MONITORED FOR BIODIVERSITY

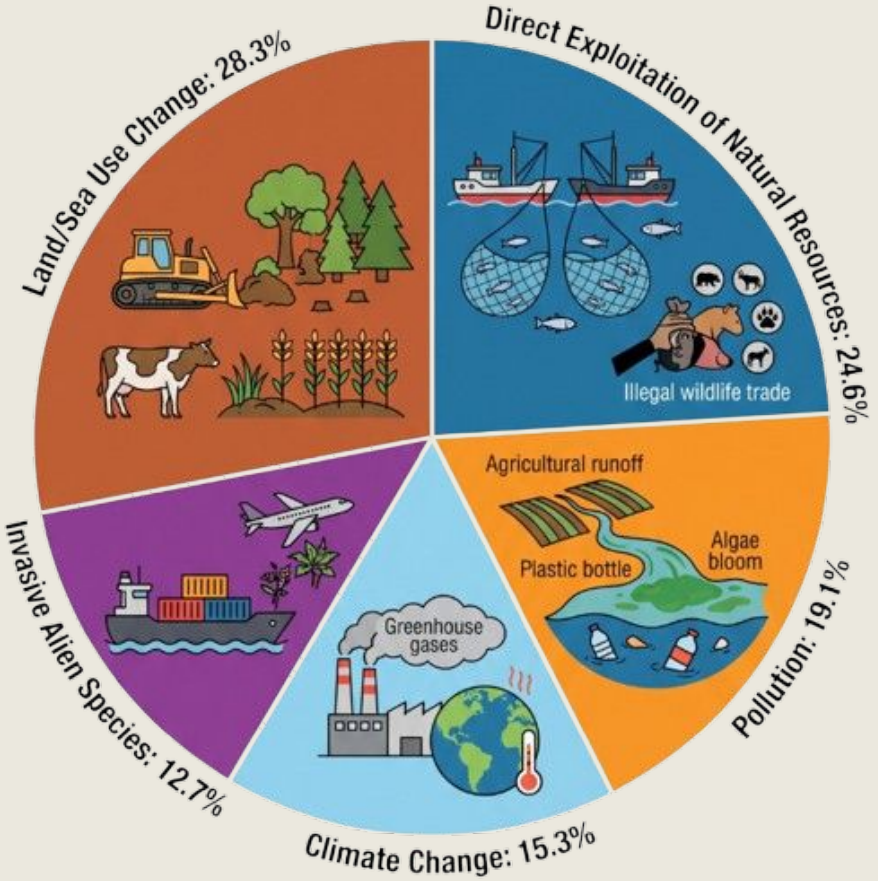
Essential Biodiversity Variables (EBVs):



*Infographic generated with Notebook LM. Percentage values verified and sourced from Jaureguiberry et al. *The direct drivers of recent Global Anthropogenic Biodiversity loss*

5 Drivers of The Biodiversity Crisis

Values derived from a meta-analysis of 45,162 studies narrowed down to 154 through extensive screening.



A Leading Driver: **Land/Sea Use Change**

- **Species Populations** (e.g., population abundance/extinction risk)
- **Ecosystem Structure** (e.g., habitat loss/ fragmentation)
- **Ecosystem Function** (e.g., primary productivity/ nutrient cycling)

B Leading Driver: **Climate Change**

- **Community Composition** (e.g., species reshuffling/range shifts)
- **Genetic Composition** (e.g., genetic diversity loss)

C Leading Driver: **Invasive Alien Species**

- **Species Traits** (e.g., morphological or behavioral trait shifts)

*Infographic generated with Notebook LM. Percentage values verified and sourced from the article Jaureguiberry et al. *The direct drivers of recent Global Anthropogenic Biodiversity loss*

Who is behind these direct drivers?

Human Activity:
The Indirect Drivers of Biodiversity Loss

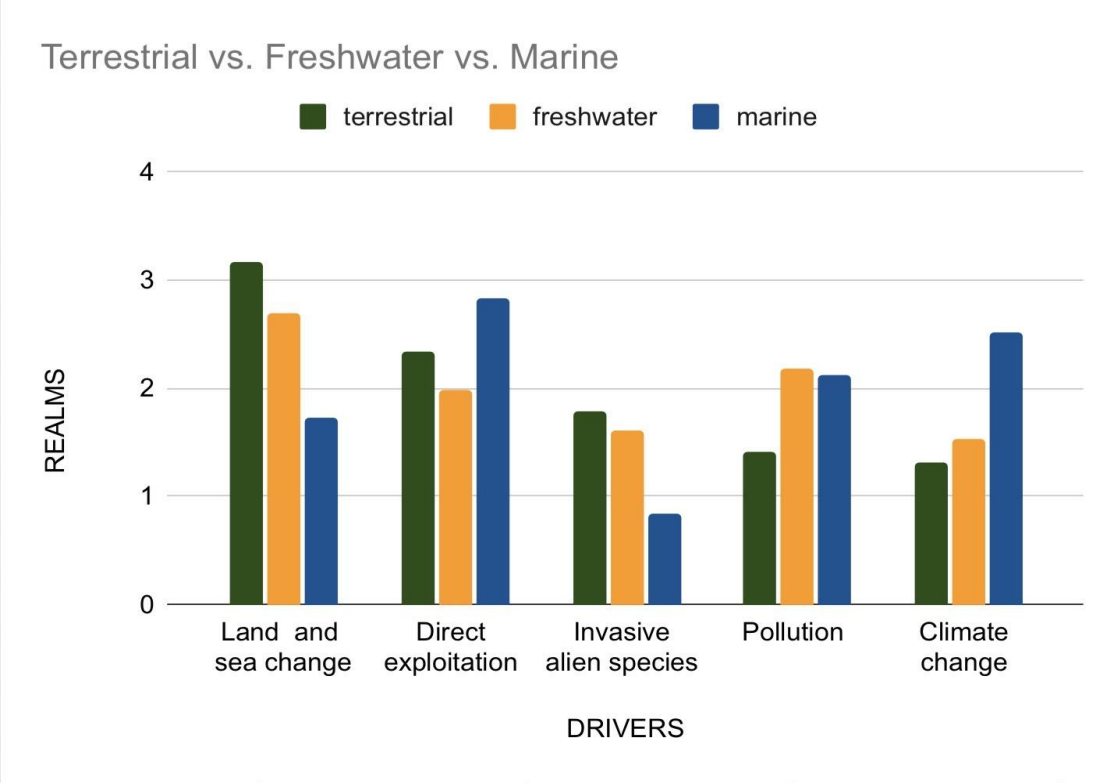


*Infographic generated Notebook LM. Percentage values verified and sourced from Jaureguiberry et al. *The direct drivers of recent Global Anthropogenic Biodiversity loss*

How are these drivers seen in different regions?

2 DIFFERENCES IN ECOSYSTEMS

HOW DID THE DRIVERS DIFFER IN EACH REALM?

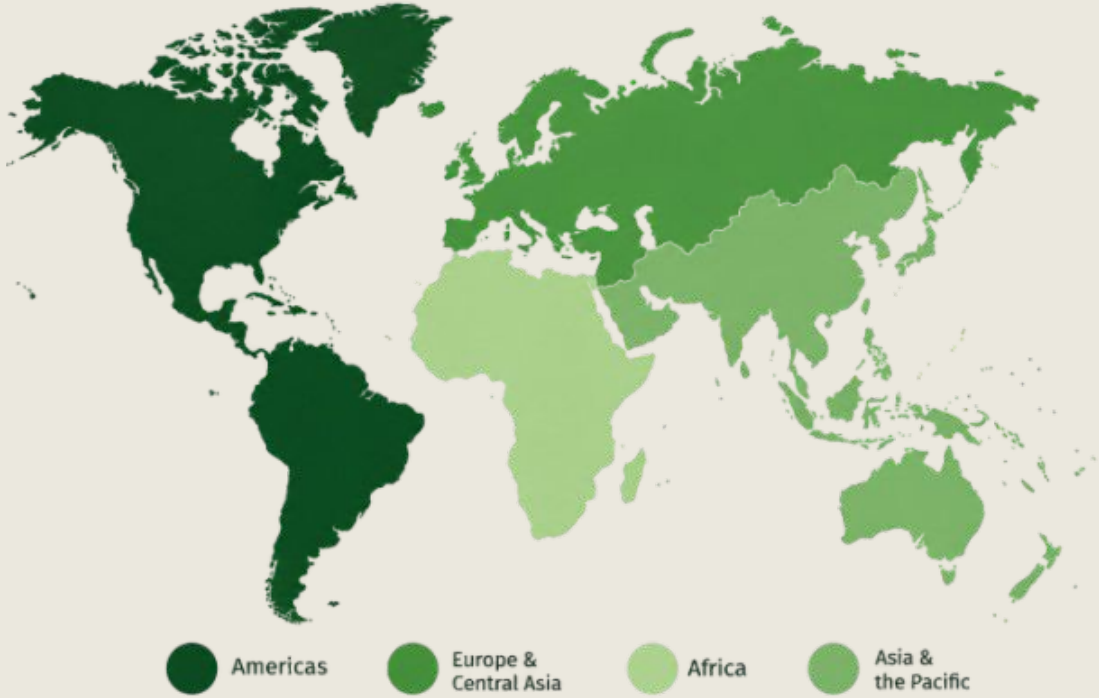


- TERRESTRIAL : Land-based environments, eg. forests, grasslands, savannas, deserts and mountains
- FRESHWATER: Rivers, lakes, streams, ponds, wetlands and freshwater marshes
- MARINE REALMS: Oceans, seas, coral reefs, estuaries, coastal systems

Data extracted from Jaureguiberry, Pedro, et al. “The Direct Drivers of Recent Global Anthropogenic Biodiversity Loss.” Science Advances, vol. 8, no. 45, Nov. 2022.

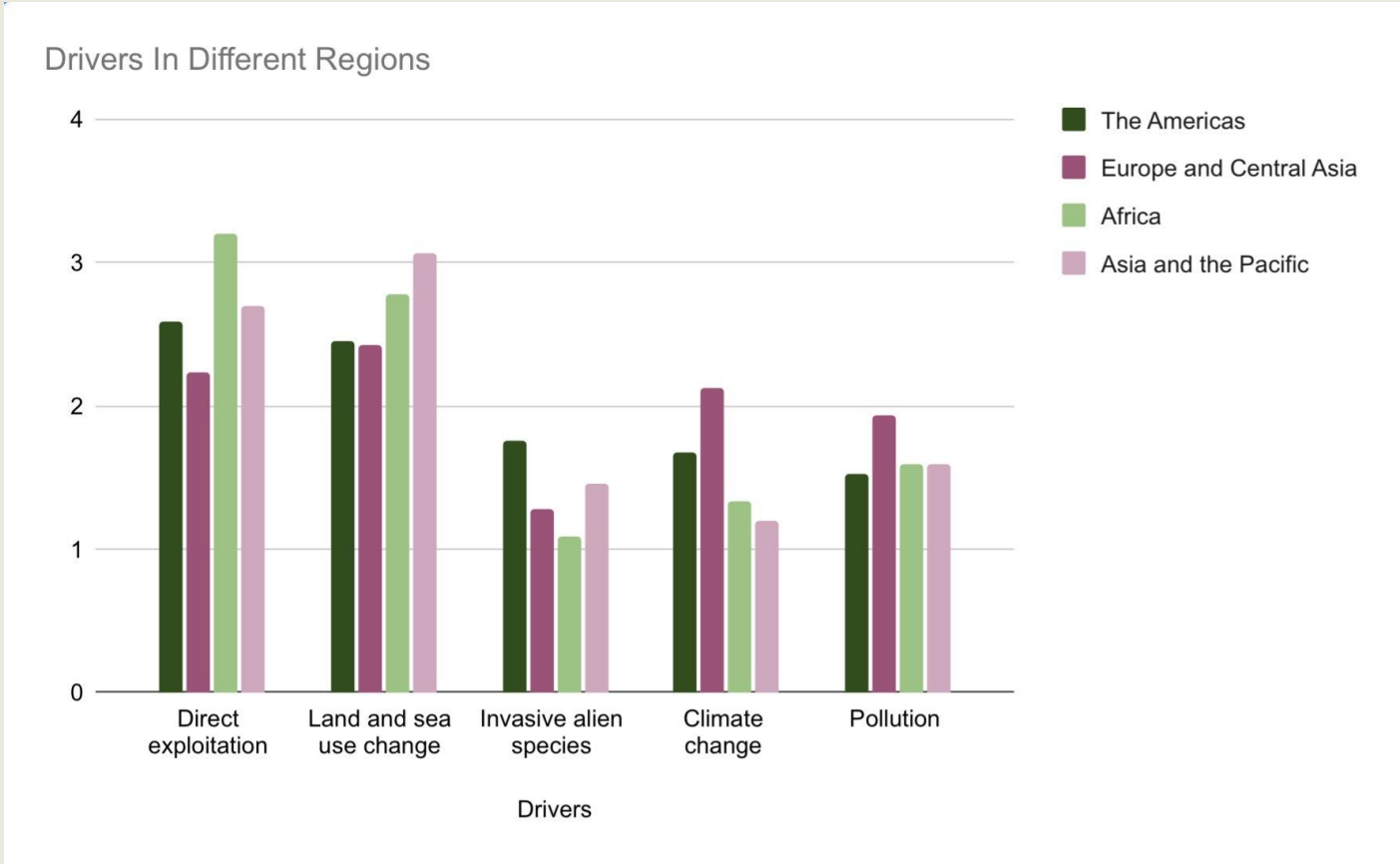
DIFFERENCES IN REGIONS

- The Americas
- Europe and Central Asia
- Africa
- Asia and the Pacific



AI-generated image

THE PREVALENCE OF EACH DRIVER IN EACH REGION



Data extracted from Jaureguiberry, Pedro, et al. “The Direct Drivers of Recent Global Anthropogenic Biodiversity Loss.” Science Advances, vol. 8, no. 45, Nov. 2022.

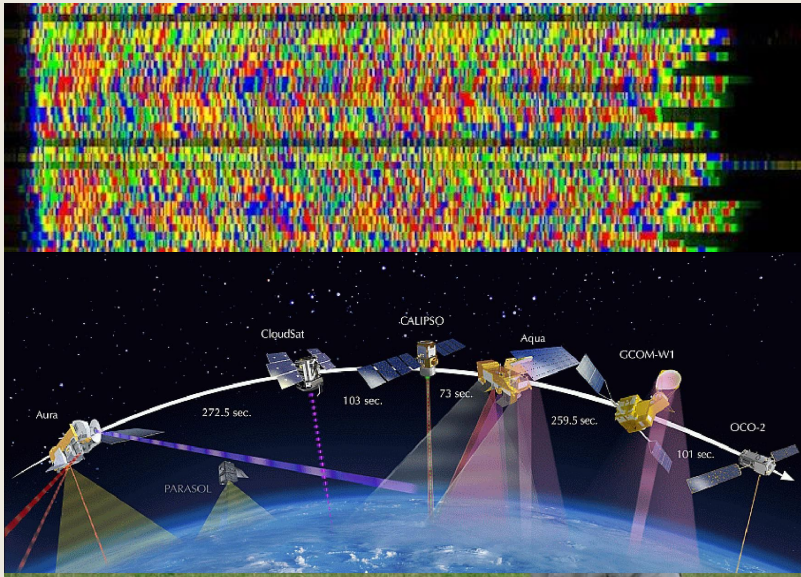
What is Biomass and how do we know how much of it there is?

3. Biomass article: WHAT IS BIOMASS AND HOW DO WE KNOW HOW MUCH OF IT IS THERE?

- **A biomass census is a comprehensive, scientific audit or quantitative assessment designed to calculate the total mass of all living biological organisms within a given scope**

- **METHODOLOGY:** Combine existing studies, DNA sequencing, satellite data, biomass sampling and environmental measurements to estimate global biomass.

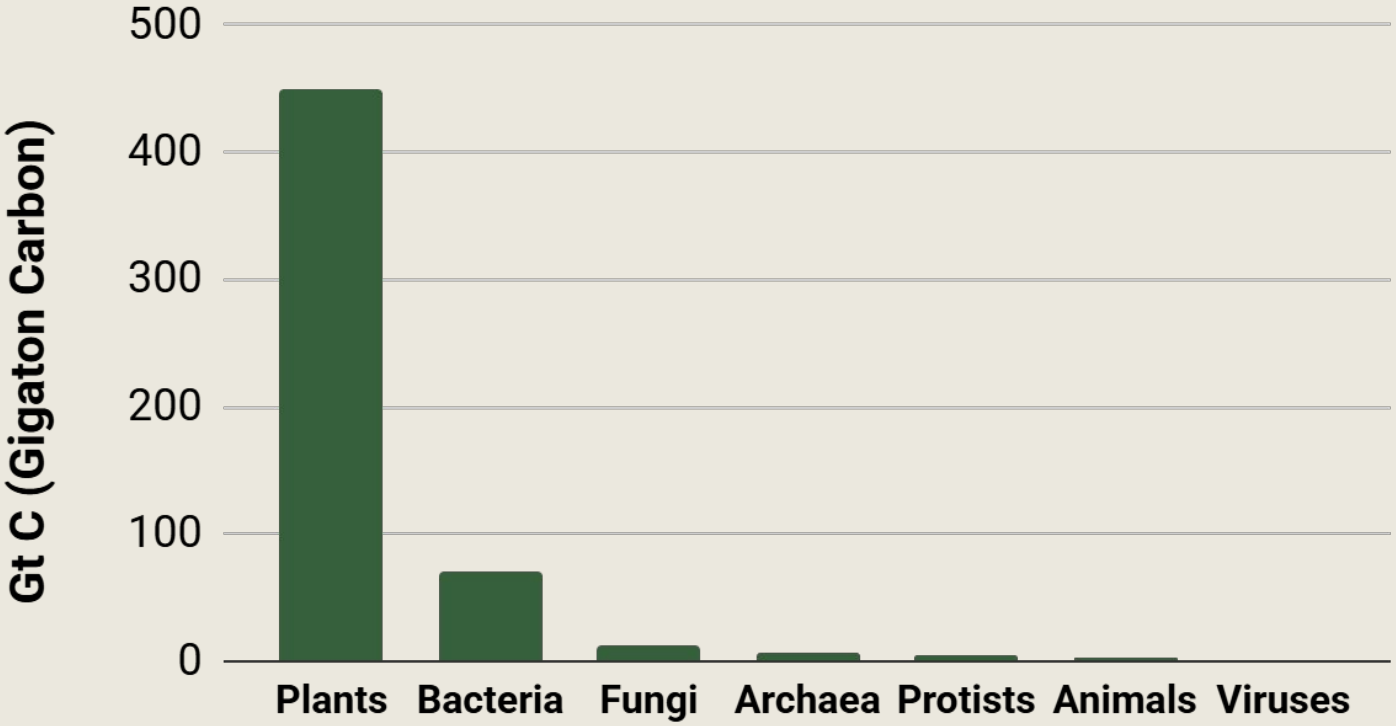
- **MEASUREMENT:** Biomass is measured in **gigatons of carbon (Gt C)**, excluding water content. **1 Gt C = 10^{15} g carbon**, linking estimates to the carbon cycle.



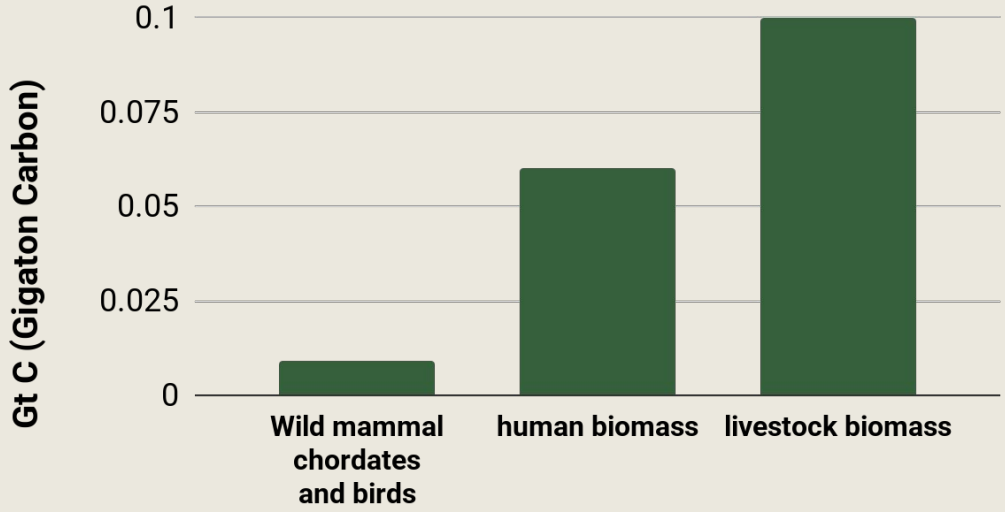


How is the biomass divided?

Biomass per taxonomy other animals



wild vs domestic mammal and human biomass



Group	Estimated Biomass (Gt C)
Plants	450
Bacteria	70
Fungi	12
archaea	7
Protists	4
animals	2
Viruses	0.2

Group	*Estimated Biomass (Gt C)*
Terrestrial arthropods	0.2
Marine arthropods	1.0
Fish chordates	0.7
Livestock chordates	0.1
Human chordates	0.06
Wild mammal chordates	0.007
Wild bird chordates	0.002

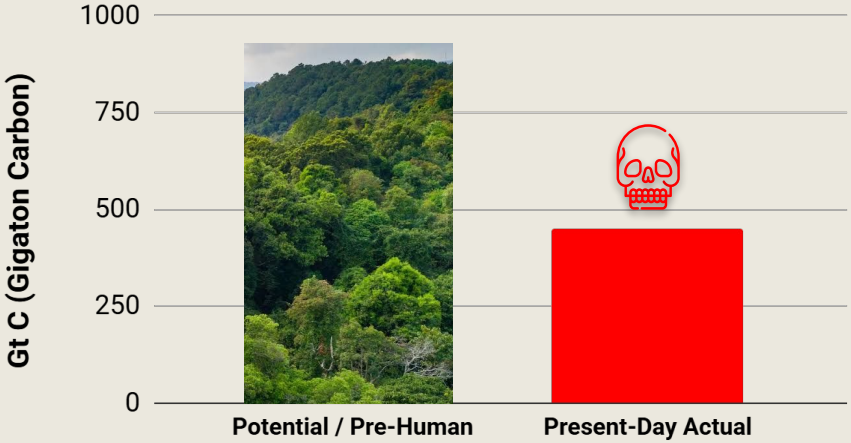
Bar-On, Yinon M., et al. “The Biomass Distribution on Earth.” Proceedings of the National Academy of Sciences, vol. 115, no. 25, 2018, pp. 6506–6511. https://doi.org/10.1073/pnas.1711842115.

What are the effects of humans of biomass?

Wild mammals per taxonomy and combined before and after human civilization



Global Plant Biomass



Global Vegetation (Plants): ~50% Total Reduction

Wild Mammals (Land & sea) sixfold Overall Collapse: Total wild mammalian biomass has fallen by ~82.5%

Wild Land Mammals: 85% reduction (7-fold drop)

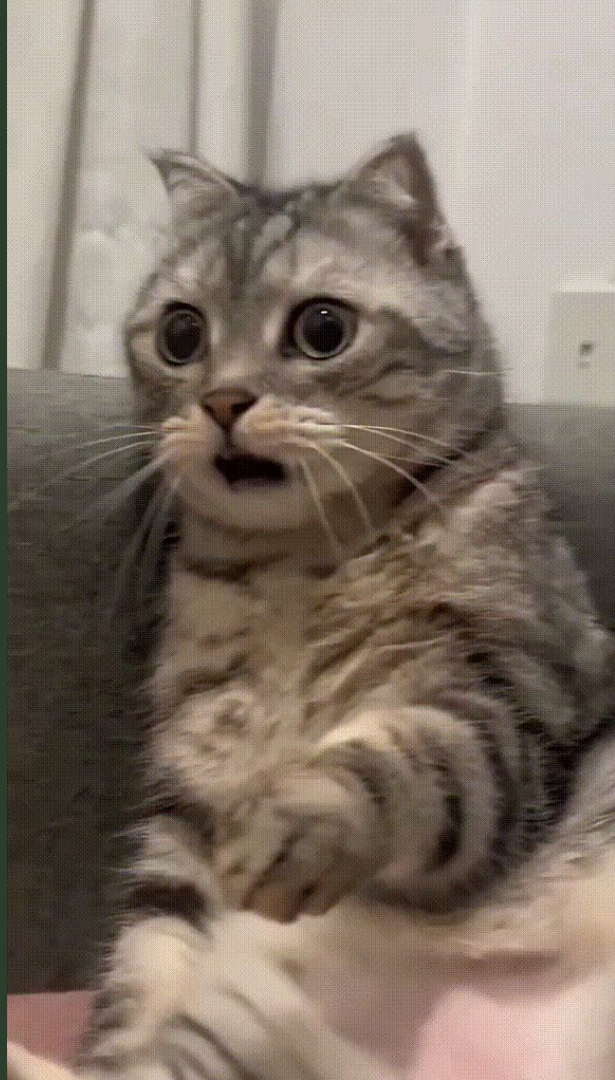
Wild Marine Mammals: 80% reduction (5-fold drop)

*note there is only limited information on the biomass regarding these specific taxa (plants, mammals (land and sea), and fish). There is no data available for the other categories (because of Lack of Historical Proxies and risk of high error margin)

Bar-On, Yinon M., et al. “The Biomass Distribution on Earth.” Proceedings of the National Academy of Sciences, vol. 115, no. 25, 2018, pp. 6506–6511. https://doi.org/10.1073/pnas.1711842115.
Bar-On, Yinon M., et al. “The Biomass Distribution on Earth.” Proceedings of the National Academy of Sciences, vol. 115, no. 25, 2018, pp. 6506–6511. https://doi.org/10.1073/pnas.1711842115.

Conclusion!

What the hell were you talking about all this time?



Source: Youtube.com/@TheDodo

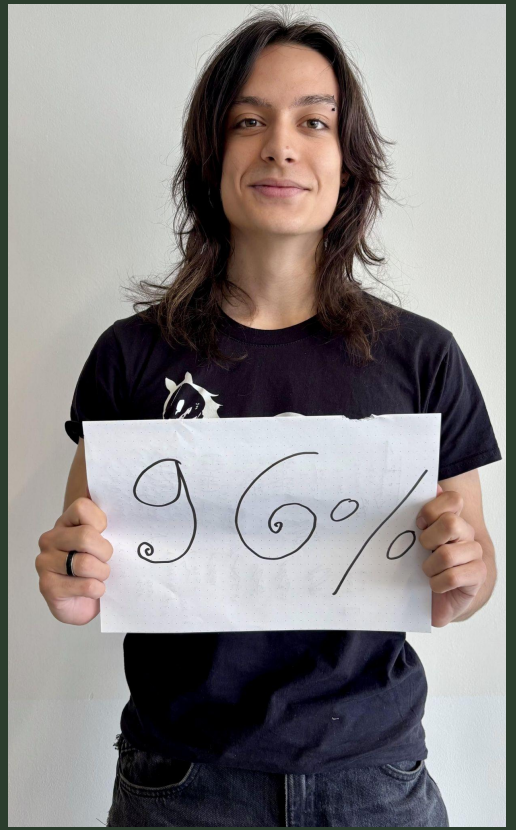
What Does "Biodiversity Crisis" Actually Mean?



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Source: Bar-On, Phillips, & Milo (2018), PNAS

THE BIODIVERSITY CRISIS & THE PATH FORWARD

01 / MAIN CAUSES



- **Land-Use Change:** Clearing ecosystems for agriculture and grazing.
- **Direct Exploitation:** Industrial overfishing and commercial logging.

02 / THE BIG MISTAKE



- **Climate-Only Focus:** Ignoring that habitat destruction is the #1 driver of loss.
- **Neglecting Habitat:** Land-hungry plans (like biofuels) often make it worse.

03 / THE PATH FORWARD

Is there light at the end of the tunnel?



- **No Shortcuts:** Reversing this crisis requires global cooperation.
- **Protect Nature First:** Guard ecosystems, end overharvesting, and save species alongside climate.

Source: Jaureguiberry et al. (2022), Science Advances

Worst-Case Scenario (Technically Accurate)



Source: Trust me, I graphed it!



We took this picture when we still had hope!

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